

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T29 · Sequences

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Sequences

F01

Recover arithmetic-series parameters

10 Q | 50 marks

F02

Use an arithmetic nth term

7 Q | 21 marks

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

$a = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Question 01

4MA1-1H Q1

MODULAR UNIT 2

3 marks

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

.....
(1)

Question 11

4MA1-1H Q24

MODULAR UNIT 2

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

Question 15

4MA1-1H Q19 demand

MODULAR UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x =$

Question 16

4MA1-1H Q20

MODULAR UNIT 2

6 marks

20 The sum of the first 80 terms of an arithmetic series, S , is 470

The 75th term of S is 14.5

The sum of the first X terms of S is 171

Work out the value of X

Show your working clearly.

$X =$

Question 12

4MA1-1H Q19 M01

MODULAR UNIT 2

5 marks

- 19** An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

.....

Question 01

4MA1-1H Q1 Q1(a)

MODULAR UNIT 2

2 marks

- 1 Here are the first five terms of an arithmetic sequence.

1 5 9 13 17

- (a) Find an expression, in terms of n , for the n th term of this sequence.

(2)

Question 02

4MA1-1H Q1 Q1(b)

MODULAR UNIT 2

1 marks

The n th term of another arithmetic sequence is $3n + 5$

(b) Find an expression, in terms of m , for the $(2m)$ th term of this sequence.

.....
(1)

Question 38

4MA1-2H Q26

MODULAR UNIT 2

5 marks

- 26 An arithmetic series has first term a and common difference d , where d is a prime number.

The sum of the first n terms of the series is S_n and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of d and the value of m

Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ k

Work out the value of k .

 $k =$

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

$a =$

85 79 73 67

- Find the value of N .
Show clear algebraic working.

$N =$

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

calculate the sum of the first 30 terms of the series.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

- The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.

(Total for Question 22 is 4 marks)

